Linguistic Phylogenetics of Ancient Nusantaran Texts: A Computational Framework for Reconstructing Pre-Indic Language Strata

Mukhlis Amien

Lab Data Sains, Universitas Bhinneka Nusantara, Indonesia

amien@ubhinus.ac.id

[DRAFT — Paper 6 of VOLCARCH Series — NOT FOR DISTRIBUTION]

> *Series context: This is Paper 6 of the VOLCARCH series. Papers 1-2 established physical taphonomic frameworks (volcanic burial; coastal submersion). Paper 3 developed a predictive ML settlement model. Paper 4 analyzed competitive inter-polity dynamics. Paper 5 introduced cosmological taphonomy, arguing that imported religious frameworks systematically obscured indigenous maritime cosmology. The present paper operationalizes the cosmological stratigraphy methodology of Paper 5 using computational linguistics, proposing a quantitative framework for measuring linguistic distance between ancient Nusantaran texts and reconstructing the pre-Indic language strata that underlie them.*

Abstract

Ancient Nusantaran texts — prasasti (stone inscriptions), kakawin (Sanskrit-Javanese epics), kidung (vernacular poetry), and the Lontara corpus of South Sulawesi — represent a stratified linguistic record spanning more than thirteen centuries. These texts vary dramatically in their degree of Sanskritic influence, their retention of Proto-Austronesian (PAn) lexical and morphological features, and their orientation toward maritime versus inland-agrarian semantic fields. Despite this variation, no systematic computational analysis of linguistic distance across this corpus has been attempted. This paper proposes a multi-feature hierarchical clustering framework for ancient Nusantaran texts, treating the corpus as a phylogenetic problem analogous to biological species classification. We define a feature vector comprising six dimensions: Sanskrit loanword density, PAn lexical retention (measured against the 200-item Swadesh list reconstructed for PAn), maritime semantic field richness, morphological complexity (with emphasis on focus-system and reduplication retention as PAn diagnostics), syntactic verb-initiality, and ritual vocabulary origin profile. We describe corpus construction methodology, propose a distance metric combining Euclidean feature distance with edit-distance on phonological correspondences, and outline a hierarchical clustering pipeline producing a dendrogram of textual-linguistic relationships. The expected output is not merely a classification of texts but a temporal stratigraphy of linguistic layers — a computational analog of the geological and cosmological stratigraphy developed in earlier papers of this series. We further propose that the dendrogram's deepest splits, corresponding to maximum linguistic distance from the Sanskrit-influenced mainstream, identify the texts most likely to preserve pre-Indic indigenous content — providing a quantitative method for prioritizing cosmological stratigraphy analysis as proposed in Paper 5.

Keywords: computational linguistics; linguistic phylogenetics; hierarchical clustering; ancient Javanese; Kawi; Proto-Austronesian; Sanskrit loanwords; Swadesh list; corpus linguistics; Nusantara; textual stratigraphy; NLP

1. Introduction

The VOLCARCH series has progressively documented four mechanisms through which Nusantaran civilization has been rendered invisible: volcanic burial of material remains (Paper 1), sea-level rise submerging coastal settlements (Paper 2), historiographic bias obscuring institutional complexity (Paper 4), and cosmological displacement erasing indigenous knowledge systems (Paper 5). Each mechanism has been addressed through its appropriate methodology: geoarchaeology, sea-level reconstruction, historical analysis, and ritual studies respectively.

Paper 5 proposed a subtraction methodology — Ajaran Asli = Contemporary Practice minus Imported Layers — as a framework for recovering pre-Indic cosmological residues. That methodology was applied qualitatively to case studies in ritual, calendrical, and epic traditions. The present paper operationalizes the same logic computationally, applying it to the textual corpus of ancient Nusantara.

The core analogy is phylogenetic. In evolutionary biology, the degree of similarity between species' DNA sequences measures their evolutionary distance from a common ancestor: organisms that diverged recently share more sequence similarity than those that diverged long ago. Applied to texts and languages, the same logic holds: texts that share a recent common linguistic tradition will share more lexical, morphological, and syntactic features than texts from traditions that diverged earlier. By measuring these distances systematically and applying hierarchical clustering, we can reconstruct a dendrogram — a tree of textual-linguistic relationships — that reveals the underlying structure of the corpus.

The critical interpretive step is the directionality of distance. In the Nusantaran corpus, the dominant direction of influence over the documented historical period runs from Sanskrit-Indic toward indigenous Austronesian: texts further from the Sanskrit pole and closer to the reconstructed Proto-Austronesian baseline are, by hypothesis, either older, less elite, less court-influenced, or produced in traditions with less exposure to Indianization. These are the texts most likely to preserve pre-Indic indigenous content — the cosmological strata whose recovery is the goal of Paper 5's methodology.

2. The Problem: No Standardized Digital Corpus

2.1 Current state of digitization

Ancient Nusantaran texts exist in varying states of digital availability. Stone inscriptions (prasasti) are partially documented in the EFEO (École française d'Extrême-Orient) database and scattered journal publications, but no unified machine-readable corpus exists. The major kakawin texts (Arjunawiwaha, Negarakertagama, Sutasoma, Ramayana Kakawin) have been edited and translated by Dutch and Indonesian scholars but are not available in standardized digital formats suitable for NLP processing. The Lontara corpus of the Bugis — including La Galigo fragments — exists in manuscript form at Leiden University Library and KITLV, with limited digital transcription.

This absence of a standardized corpus is itself a contribution opportunity. Before the analytical pipeline proposed in this paper can be executed, a corpus construction effort is required. Section 3 describes this effort as the first phase of the research program.

2.2 Challenges specific to ancient Nusantaran NLP

Beyond corpus availability, ancient Nusantaran texts present several NLP challenges that distinguish them from better-studied ancient language corpora such as Classical Latin or Ancient Greek. First, the script systems are diverse: Old Javanese (Kawi) texts use a Brahmi-derived abugida script that was not designed for the phonological features of

Austronesian languages and that encodes vowel length and consonant clusters in ways that require specialized preprocessing. Lontara script is an abugida of different design with its own encoding challenges. Romanized transliterations exist for most major texts but vary in convention across editions.

Second, the language situation is diglossic and internally variable. Kawi texts mix Old Javanese vernacular with Sanskrit borrowings at rates ranging from under 10% (in older, more vernacular texts) to over 60% (in the most heavily Sanskritized kakawin). Identifying and tagging Sanskrit borrowings requires a Sanskrit lexicon cross-referenced against Old Javanese dictionaries — a resource that exists in incomplete form (Zoetmulder 1982) but has not been digitized for NLP use.

Third, morphological analysis is complex. Old Javanese uses a rich system of prefixes, infixes (notably -um- and -in-, which are diagnostic PAn features), and suffixes, plus Sanskrit-derived nominal compounds. Accurate morphological segmentation is prerequisite for most downstream linguistic analysis. The author's previous work on ModernKataKupas (a morphological segmenter for Modern Indonesian) provides a relevant methodological foundation, though the morphological system of Old Javanese differs substantially from Modern Indonesian and requires adapted approaches.

3. Corpus Construction

3.1 Target corpus: primary texts

The target corpus for this study comprises texts distributed across the full temporal and linguistic range of documented ancient Nusantaran writing, selected to maximize coverage of the Sanskrit-to-PAn continuum. Table 1 presents the proposed corpus.

Table 1: Proposed corpus of ancient Nusantaran texts

Text	Date (CE)	Language	Genre	Expected Sanskrit density
Prasasti Kedukan Bukit	683	Old Malay	Royal inscription	Low (~15%)
Prasasti Canggal	732	Sanskrit/OJ	Royal inscription	Very high (~80%)
Arjunawiwaha	~1035	Old Javanese (Kawi)	Kakawin (epic)	High (~55%)
Negarakertagama	1365	Old Javanese (Kawi)	Kakawin (chronicle)	High (~50%)
Sutasoma	~1380	Old Javanese (Kawi)	Kakawin (philosophical)	High (~60%)
Pararaton	~1400-1600	Middle Javanese	Chronicle/prose	Medium (~30%)
Kidung Sunda	~1400s	Middle Javanese	Kidung (vernacular epic)	Low-medium (~20%)

Calon Arang	~1400s	Old Javanese/Bali	Prose narrative	Medium (~35%)
La Galigo (selected passages)	~1300-1600?	Old Bugis (Lontara)	Epic/cosmological	Very low (~5%)
Prasasti Yupa (Kutai)	~400	Sanskrit	Votive inscription	Near 100%

3.2 Corpus preprocessing pipeline

Corpus construction requires a standardized preprocessing pipeline converting heterogeneous source materials into a uniform format suitable for feature extraction. The proposed pipeline comprises five stages:

Stage 1: Transliteration standardization

All texts will be converted to a standardized Roman transliteration following the conventions of Zoetmulder (1982) for Old Javanese and Matthes (1874) for Bugis/Lontara. Where multiple editions exist, the most recent scholarly edition will be used as the base text, with variant readings recorded in a parallel annotation layer.

Stage 2: Tokenization and morphological segmentation

Tokenization of Old Javanese requires handling of Sanskrit compound words (samasa) which may span what would be whitespace-separated tokens in modern text. A specialized tokenizer will be developed, building on ModernKataKupas architecture but extending it to handle: (a) Sanskrit nominal compounds with internal sandhi; (b) Old Javanese infix insertion (-um-, -in-) which modifies the phonological form of the root; (c) Lontara script's handling of final consonants (which are systematically elided in the script and must be reconstructed for morphological analysis).

Stage 3: Lexical tagging — Sanskrit vs. indigenous

Each token will be tagged for its lexical origin: Sanskrit borrowing, Proto-Austronesian-derived, or indeterminate. Sanskrit tagging uses a lookup against a digitized version of Monier-Williams (1899) cross-referenced with Zoetmulder (1982). PAn-derived tagging uses the ACD (Austronesian Comparative Dictionary, Blust and Trussel 2010, online) reconstructed forms. Tokens matching neither source are tagged as indeterminate, constituting a category of interest for the pre-Indic indigenous stratum.

Stage 4: Semantic field annotation

A subset of tokens in each text will be annotated for semantic field membership using a custom tagset developed for this project. The critical semantic fields are: MARITIME (nautical vocabulary, fish species, wave/tide/current terminology, boat-part vocabulary, navigation terms), INLAND-AGRARIAN (rice cultivation vocabulary, irrigation terms, soil types, plough technology), COSMOLOGICAL-INDIC (Sanskrit cosmological terms, Hindu-Buddhist ritual vocabulary), COSMOLOGICAL-INDIGENOUS (terms for ancestral spirits, indigenous ritual objects, indigenous spatial orientation), and POLITICAL-ADMINISTRATIVE (governance vocabulary, with sub-tagging for Sanskrit-derived vs. indigenous terms).

Stage 5: Syntactic annotation

A dependency parse will be applied to each text using an adapted version of the Universal Dependencies framework. Key syntactic features to be extracted include: verb-subject-object order (PAn languages are verb-initial; Sanskritic influence tends toward SOV or SVO order);

presence of focus-system morphology (a characteristic PAn feature largely absent from Sanskrit); and frequency of nominal vs. verbal predication.

4. Feature Vector Design

4.1 The six-dimensional feature space

Each text in the corpus will be represented as a point in a six-dimensional feature space. The six dimensions are chosen to maximize discriminative power across the Sanskrit-to-PAn continuum while remaining computable from the preprocessed corpus. Table 2 summarizes the feature vector.

Table 2: Feature vector dimensions

F	Feature	Operationalization	Range	Interpretation
F1	Sanskrit loanword density	% tokens tagged SANSKRIT	0.0 – 1.0	High = strongly Indianized
F2	PAn lexical retention	% Swadesh-200 items with PAn cognate	0.0 – 1.0	High = close to indigenous baseline
F3	Maritime semantic richness	MARITIME tokens / total tokens × diversity index	0.0 – 1.0	High = maritime-oriented text
F4	PAn morphological retention	Frequency of -um-/-in- infix + reduplication per 1000 tokens	0 – ~50	High = strong PAn morphological layer
F5	Verb-initiality index	% clauses with verb in initial position	0.0 – 1.0	High = PAn syntactic retention
F6	Indigenous ritual vocab ratio	COSMOLOGICAL-INDIGENOUS / (COSM-INDIC + COSM-INDIGENOUS)	0.0 – 1.0	High = indigenous cosmological content

4.2 Feature normalization and weighting

Before distance computation, features will be normalized to a common scale using z-score standardization across the corpus. Feature weighting is a design choice with significant implications for the resulting dendrogram. Two weighting schemes will be compared: (a) equal weighting, giving each of the six dimensions equal influence on the distance metric; and (b) theory-guided weighting, upweighting F2 (PAn lexical retention) and F3 (maritime semantic richness) on the grounds that these features are most directly relevant to the pre-Indic stratigraphy question. Both dendrograms will be reported, with divergences between them analyzed as informative about which features drive the clustering.

4.3 Incorporating the Swadesh list for Old Javanese

The 200-item Swadesh list was designed to capture the most stable and borrowing-resistant vocabulary across languages — body parts, basic kinship terms, low numbers, fundamental

natural phenomena, and universal actions. For Old Javanese, applying the Swadesh list requires identifying the Kawi equivalent of each Swadesh item and determining whether that equivalent is PAn-derived or Sanskrit-borrowed.

This task has not been previously undertaken systematically. Zoetmulder (1982) provides the Old Javanese lexicon; the ACD (Blust and Trussel 2010) provides PAn reconstructions with Javanese reflexes where documented. Mapping these resources against the Swadesh-200 list will generate a binary retention vector for each text: for each of the 200 Swadesh items, is the Kawi equivalent PAn-derived (1) or Sanskrit-borrowed (0)? The F2 score is the mean of this vector. A text with F2 = 0.80 retains 80% of its basic vocabulary from the PAn substrate despite Sanskrit borrowing pressure — a strong indicator of indigenous linguistic continuity.

5. Distance Metric and Clustering Algorithm

5.1 The composite distance metric

The distance between two texts i and j in the feature space is computed as a weighted combination of two components: Euclidean distance in the six-dimensional feature space (capturing overall linguistic profile similarity) and a phonological correspondence distance (capturing the degree to which the texts share systematic sound correspondences, as expected for texts within the same scribal or oral tradition).

The composite distance metric is:

> $d(i,j) = \alpha \cdot d_Euclidean(F_i, F_j) + (1-\alpha) \cdot d_phonological(i,j)$

where α is a mixing parameter (default 0.7, treating feature profile as primary and phonological correspondence as secondary), d_Euclidean is the normalized Euclidean distance in feature space, and d_phonological is derived from a phonological correspondence matrix computed from shared vocabulary items.

5.2 Phonological correspondence distance

Phonological distance between texts is computed using the following procedure. For each pair of texts, identify the set of shared vocabulary items (tokens appearing in both texts that are tagged as PAn-derived in at least one text). For each shared item pair, compute the edit distance on the phonological representation. Average across all shared items to obtain a mean phonological distance. This measure captures systematic sound correspondences: texts within the same oral or scribal tradition will share regular sound correspondences, while texts from distinct traditions will show irregular or absent correspondences even for cognate items.

5.3 Hierarchical clustering and dendrogram construction

Hierarchical agglomerative clustering will be applied to the distance matrix using Ward's linkage criterion, which minimizes the total within-cluster variance at each merge step. Ward's linkage is preferred over single or complete linkage for this application because it tends to produce compact, well-defined clusters that are more interpretable as linguistic families or traditions.

The output dendrogram will be validated using two standard metrics: cophenetic correlation (measuring how faithfully the dendrogram preserves the original pairwise distances) and silhouette score (measuring cluster cohesion and separation at each level of the hierarchy).

Bootstrap resampling of the feature vectors (n=1000) will be used to assess the stability of individual splits in the dendrogram.

6. Expected Results and Interpretation

6.1 Hypothesized dendrogram structure

Based on the known properties of the corpus, we hypothesize the following broad structure in the resulting dendrogram. The most distant split in the tree will separate La Galigo from all other texts: its near-zero Sanskrit density, high PAn retention, and predominantly maritime semantic field should place it at maximum distance from the Kawi kakawin cluster. This split, if confirmed, provides direct quantitative support for Paper 5's argument that La Galigo preserves a pre-Indic cosmological tradition qualitatively distinct from the Javanese-Sanskrit mainstream.

Within the Kawi corpus, we expect a secondary split between the most heavily Sanskritized texts (Prasasti Canggal, Sutasoma, Arjunawiwaha) and the more vernacular texts (Pararaton, Kidung Sunda). The prasasti will cluster differently from the kakawin despite similar dates, reflecting the genre difference between official royal epigraphy (maximizing Sanskrit markers of legitimacy) and narrative or poetic literature (retaining more vernacular features). The Old Malay texts (Prasasti Kedukan Bukit and related Sriwijayan inscriptions) will likely form a distinct cluster from the Javanese texts, reflecting the independent Malay linguistic tradition.

6.2 The pre-Indic stratum indicator

The most interpretively significant output of the analysis is the identification of texts with high F6 (indigenous ritual vocabulary ratio) combined with high F2 (PAn lexical retention) and high F3 (maritime semantic richness). These texts sit at the maximum distance from the Sanskrit pole in the feature space and are by hypothesis the richest sources for pre-Indic cosmological content.

The working hypothesis, subject to empirical test, is that this cluster will include La Galigo, the Kidung genre texts, certain Balinese lontar manuscripts not included in the main Kawi corpus, and possibly some prasasti from regions with less intensive Brahmanical influence (eastern Java periphery, Borneo, outer islands). These texts would constitute a priority list for the qualitative cosmological stratigraphy analysis of Paper 5 — a quantitatively-derived roadmap for recovering indigenous intellectual heritage.

6.3 The indeterminate vocabulary as archaeological layer

Tokens tagged as 'indeterminate' — neither Sanskrit nor PAn-derived — constitute a particularly interesting category. In a well-studied language pair such as English and French, indeterminate vocabulary would be a methodological failure. In the Nusantaran corpus, it may represent genuine pre-Indic indigenous vocabulary that has no documented PAn reconstruction because PAn reconstruction is itself incomplete.

The distribution of indeterminate vocabulary across semantic fields is therefore analytically significant. If indeterminate tokens cluster disproportionately in maritime, ritual-indigenous, and cosmological-indigenous semantic fields — precisely the fields where pre-Indic content is hypothesized to survive — this is convergent evidence for a layer of indigenous technical and cosmological vocabulary that predates both Sanskrit contact and the documented PAn reconstruction. This vocabulary layer would be an archaeological find in the strict sense: material that was present but invisible until the right analytical method exposed it.

7. Connection to ModernKataKupas and Future NLP Infrastructure

7.1 Extending ModernKataKupas to Old Javanese

ModernKataKupas, the author's open-source morphological segmenter for Modern Indonesian (Amien, 2023), operates on the principle that Indonesian words can be decomposed into root morphemes plus affixes using a rule-based system informed by linguistic knowledge of Indonesian morphophonology. Old Javanese morphology shares several structural features with Modern Indonesian — both are Austronesian languages with productive affixation — but differs in ways that require systematic adaptation.

The most significant differences are: (1) Old Javanese infix -um- and -in- are more productive and morphophonologically complex than their Modern Indonesian cognates; (2) Sanskrit nominal compounds in Old Javanese create multi-morpheme tokens with internal sandhi that must be analyzed differently from native word formation; (3) the lexicon is substantially different, requiring a dedicated Old Javanese root lexicon rather than extension of the Modern Indonesian one. A modified version of ModernKataKupas, tentatively named KawiKupas, is proposed as a deliverable of this research program.

7.2 Toward an open corpus for ancient Nusantaran NLP

The corpus construction effort described in Section 3 will produce, as a byproduct, the first standardized open digital corpus of ancient Nusantaran texts in a format suitable for NLP research. This corpus — with standardized transliteration, morphological segmentation, lexical tagging, semantic field annotation, and syntactic annotation — would be a research infrastructure contribution independent of the specific analytical results of this paper.

The corpus will be released under an open license (Creative Commons Attribution) via the project GitHub repository, with documentation sufficient for researchers in historical linguistics, comparative religion, and Southeast Asian studies to use it without specialized NLP expertise. This addresses a significant gap in digital humanities infrastructure for Indonesian and Nusantaran studies.

8. Limitations and Caveats

The computational framework proposed here has several important limitations that must be acknowledged. First, the corpus is small by NLP standards — ten to fifteen texts comprising perhaps 500,000 total tokens. Statistical power is limited, and feature estimates for shorter texts will carry high uncertainty. The bootstrap validation procedure is designed to quantify this uncertainty, but it cannot eliminate it.

Second, the Sanskrit loanword tagging assumes that Sanskrit borrowings can be clearly identified by lexical lookup. In practice, some Sanskrit borrowings are so deeply integrated into Kawi that they are no longer recognized as loans; conversely, some items that resemble Sanskrit forms may be coincidental sound similarities. The tagging will carry an estimated error rate of 5-15% based on the completeness of available lexical resources.

Third, the PAn reconstruction used as the baseline (ACD, Blust and Trussel 2010) is itself a scholarly reconstruction with its own uncertainties. PAn is reconstructed from ~1,200 daughter languages using the comparative method; the reconstruction of specific items

varies in confidence. Using PAn as a baseline introduces these uncertainties into the F2 feature.

Fourth and most fundamentally, the dendrogram produced by this analysis is a classification of texts, not a direct reconstruction of the pre-Indic language stratum. The inference from 'this text has high PAn retention and low Sanskrit density' to 'this text preserves pre-Indic cosmological content' is a hypothesis, not a logical necessity. The cosmological stratigraphy analysis of Paper 5 remains the essential complementary method for interpreting what the computational clustering reveals.

9. Conclusions

This paper has proposed a computational linguistic framework for analyzing the ancient Nusantaran textual corpus as a phylogenetic problem, using hierarchical clustering on a six-dimensional feature space to reconstruct the stratified linguistic history embedded in these texts. The framework operationalizes the cosmological stratigraphy methodology of Paper 5, providing a quantitative method for identifying which texts in the corpus are most likely to preserve pre-Indic indigenous content.

The proposed analysis requires, as prerequisites, a corpus construction effort that will itself constitute a research infrastructure contribution: the first standardized open digital corpus of ancient Nusantaran texts for NLP research, accompanied by KawiKupas, an adapted morphological segmenter for Old Javanese. These tools will serve researchers across historical linguistics, comparative religion, digital humanities, and Southeast Asian studies far beyond the specific applications of this paper.

The deepest aspiration of this paper, and of the VOLCARCH series as a whole, is to demonstrate that the apparent thinness of the Nusantaran civilizational record is an artifact of the analytical methods applied to it — not a feature of the civilization itself. Sediment buries temples. Sea water drowns coastal settlements. Historiographic bias ignores what does not fit the template. Cosmological displacement reframes indigenous knowledge as superstition. And the absence of a standardized computational corpus leaves a literary tradition of extraordinary richness — including the longest epic in human history — invisible to the quantitative tools of modern scholarship. This paper is an attempt to remove one more layer of that invisibility.

Draft Notes & Placeholders

[ ] Section 3.2, Stage 3 — Sanskrit tagging pipeline needs validation against a held-out test set with known etymology. Consider using Negarakertagama as the validation text (best-studied text in the corpus).

[ ] Section 4.3 — The Swadesh-200 to Old Javanese mapping is the most labor-intensive component of the project. Estimate: 40-60 hours of expert linguistic annotation. Needs a collaborator with Old Javanese expertise — possibly UNAIR or UGM linguistics department.

[ ] Section 5.1 — The alpha mixing parameter (0.7/0.3 split between feature and phonological distance) is arbitrary at this stage. Should be validated empirically or justified theoretically in final version.

[ ] Section 6.1 — The hypothesized dendrogram structure should be presented more cautiously — these are predictions, not findings. Reframe as explicit hypotheses H1-H4 for clarity.

[ ] Section 7.1 — KawiKupas as a named tool — check whether this name is already in use in the literature before committing to it.

[ ] La Galigo corpus access: contact Universiteitsbibliotheek Leiden directly. Dr. Roger Tol (KITLV) and Prof. Christian Pelras are key contacts for Bugis manuscript access. Check current status of Lontara digitization projects.

[ ] Consider adding a Section 10: Roadmap, specifying which components are (a) immediately executable with existing resources, (b) require corpus construction first, (c) require external collaboration.

[ ] Journal target: Computational Linguistics, Language Resources and Evaluation, or Digital Scholarship in the Humanities. This paper is the most straightforwardly publishable of the series in terms of technical novelty.

[ ] Connection to existing work: check Rajeg et al. (2021) on computational approaches to Balinese; check if any previous computational work on Kawi exists in Indonesian NLP literature.

References (Preliminary)

Blust, R., Trussel, S., 2010. Austronesian Comparative Dictionary. Web edition. URL: http://www.trussel2.com/acd/

Matthes, B.F., 1874. Boegineesch-Hollandsch Woordenboek. Martinus Nijhoff, The Hague.

Monier-Williams, M., 1899. A Sanskrit-English Dictionary. Clarendon Press, Oxford.

Noorduyn, J., 1955. Een achttiende-eeuwse kroniek van Wadjo. De Bussy, 's-Gravenhage.

Pelras, C., 1996. The Bugis. Blackwell, Oxford.

Prapanca, M., 1365/1995. Nagarakertagama. Trans. S. Robson. KITLV Press, Leiden.

Zoetmulder, P.J., 1982. Old Javanese-English Dictionary. 2 vols. Martinus Nijhoff, The Hague.

Amien, M., 2023. ModernKataKupas: A Rule-Based Morphological Segmenter for Modern Indonesian. [GitHub repository / previous publication]

Amien, M., 2026a–e. VOLCARCH Series, Papers 1–5. Lab Data Sains, Universitas Bhinneka Nusantara.

[Additional references to be added — especially computational linguistics and digital humanities literature on ancient language NLP]